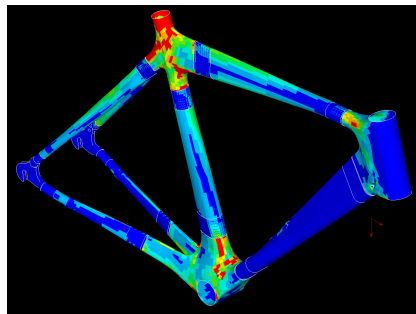
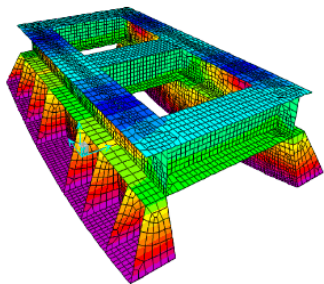




(Credit: [Form Lab](#))



(Credit: [Adesol](#))

Introduction

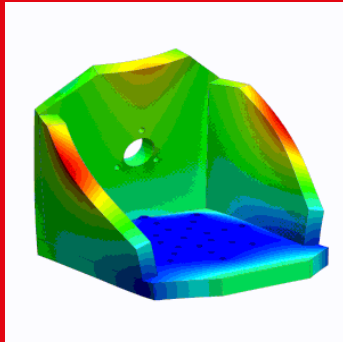
General information

ME-473

Dynamic finite element analysis of structures

Stefano Burzio

Welcome!



- Who are the instructors?
- How is the learning process structured?
- What reading materials are recommended?
- What does the curriculum include?
- How will my learning be assessed?
- Why is dynamic FEA important?

Lecturer



Stefano Burzio
Scientific collaborator

Teaching assistants



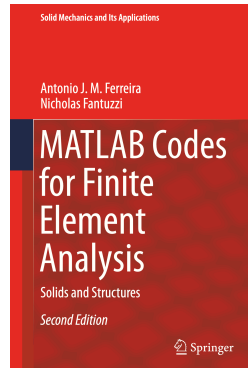
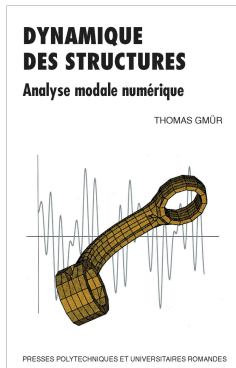
Timothée Salamon
PhD student at LFMI



Gabriel Mollet
Master's student in Mechanical Engineering

Who am I?

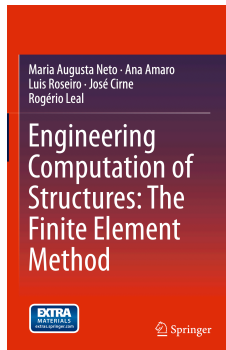
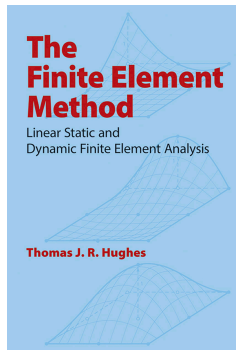




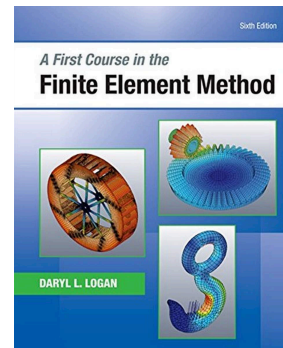
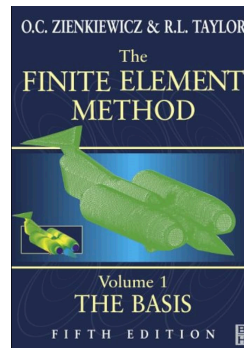
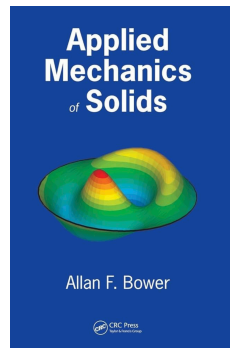
Main references:

- Thomas Gmür
Dynamique des structures (PPUR)
EPFL library: (07 534 GMU 2012)
- Antonio Ferreira and Nicholas Fantuzzi
MATLAB Codes for Finite Element Analysis
Available online ([link](#))

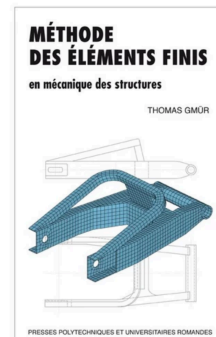
English



All-time classics



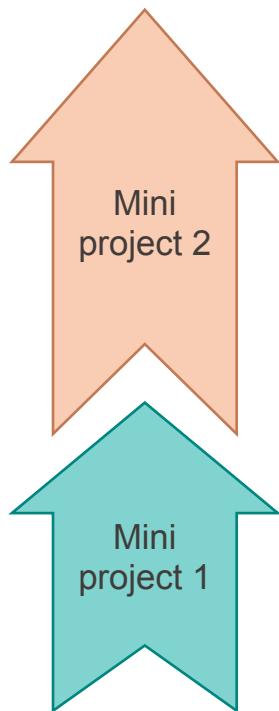
ME-372



Week	Module	Lecture topic	Problem set	Mini projects
1	Linear elastodynamics	Strong and weak forms	1	Groups formation
2		Galerkin and finite element methods	2	
		Jeûne fédéral		
3		Systematisation of the procedure	3	
4		Rotational or dissipative structures	4	
5	Special structural elements	Trusses	5	Project 1 statement
6		Beams, frames and grids	6	
7		Kirchoff plates	7	
8		Riessen-Mindlin plates	8	Project 1 submission
9	Analysis of free and forced vibrations	Lancoz, subspace algorithms	9	Project 2 statement
10		Gyroscopic and dissipative systems		
11		CADFEM presentation : Thierry Bleuler		
12		Newmark schemes	10	
13		Wilson schemes		Project 2 submission

EPFL Mini projects submission guidelines

7



- **Objective:** put course content into practice.
- **Groups:** 3 to 4 students (register groups on Moodle)
- **Group assessment:**
 - work divided equally among the group.
 - evaluation criteria on Moodle.
- **Deliverables:**
 - PDF report (explaining your analysis and detailing the code)
 - Simulation file (MATLAB, ANSYS, Abaqus)
- **Deadlines:**

Project	Topic	Statement	Submission
1	Modal analysis	October 12	November 13
2	Transient analysis	November 16	December 18

1. Understand the fundamentals of dynamic FEA

Develop a strong theoretical foundation in the principles of dynamic analysis, including *modal analysis* and *frequency response*.

2. Analyse the dynamic behaviour of structural components

Study the dynamic response of specific structures such as *rods*, *trusses*, *beams*, *plates*, and *shells* under various loading and boundary conditions.

3. Develop proficiency in finite element formulation for dynamic problems

Learn to derive and implement finite element equations for time-dependent structural dynamics problems, including damping and rotational effects.

4. Apply computational methods for solving dynamic systems

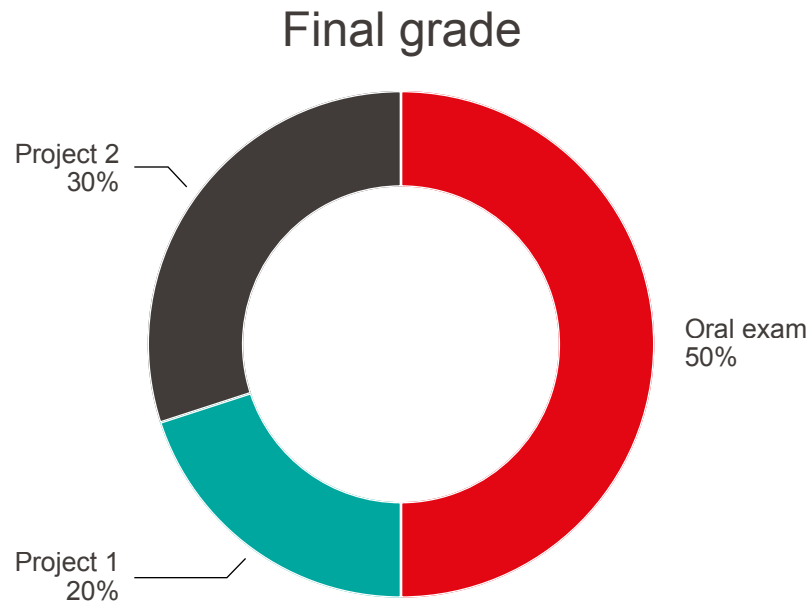
Utilise numerical techniques such as direct integration, *modal superposition*, and *time-stepping methods* to solve complex dynamic problems.

5. Implement advanced applications in engineering design

Apply dynamic FEA to real-world engineering problems, such as vibration analysis and linear dynamic response to dynamic loads, using commercial software.

Final exam

- Fall session
- Oral exam
 - Presentation of mini-project 2
 - Questions about the course



EPFL The finite element analysis dichotomy

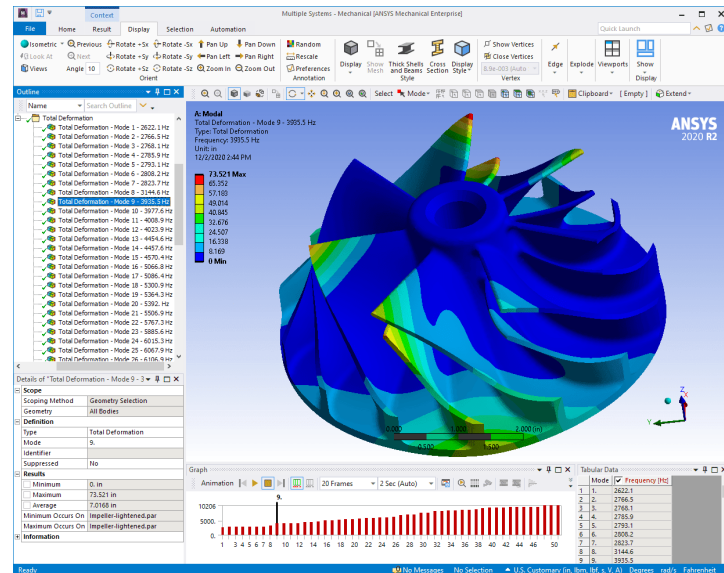
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Code developers

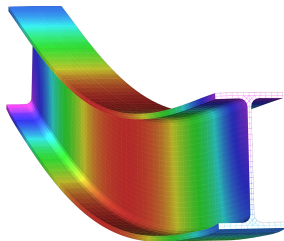
Tool users

```
35 static public void verify()
36 {
37     cgiSolverBlazeClass solver = new cgiSolverBlazeClass();
38     solver.createStructure();
39
40     StringBuilder sb = new StringBuilder();
41     solver.getDefaultTestPath(ref sb);
42     string sInputFileName = sb.ToString() + "\\tests\\newModels\\Verify-Example4-Sparse.r3a";
43
44     // the following should be called after createStructure()
45     cgiSolverBlazeClass.ListMessageDelegate listMsg = new cgiSolverBlazeClass.ListMessageDelegate(Callback);
46     cgiSolverBlazeClass.StatusMessageDelegate statusMsg = new cgiSolverBlazeClass.StatusMessageDelegate(StatusCallback);
47     cgiSolverBlazeClass.SparseSolverProgressDelegate sparseMsg = new cgiSolverBlazeClass.SparseSolverProgressDelegate(MkProgress);
48     solver.setListMessageFunction(listMsg);
49     solver.setStatusMessageFunction(statusMsg);
50     solver.setSparseSolverStatusFunction(sparseMsg);
51
52     solver.setModelType((int)cgiModelEnum.kModel_Frame2D);
53
54     // LENGTH=ft; DIMENSION=in; FORCE=kip; FORCE_LINE=kip/ft; MOMENT=kip-ft; FORCE_SURFACE=lb/ft^2;
55     // DISPLACEMENT_TRANS=in; DISPLACEMENT_ROTATE=rad; MODULUS=kip/in^2; WEIGHT_DENSITY=lb/ft^3; STRESS=lb/in^2
56     // SPRING_TRANS_1D=lb/in; SPRING_ROTATE_1D=lb-in/rad; SPRING_TRANS_2D=kip/in^2; SPRING_TRANS_3D=kip/in^3
57     solver.setStandardEnglishUnits();
58
59     // define materials
60     List<cgiMaterialCll> listMat = new List<cgiMaterialCll>();
61     cgiMaterialCll mat = new cgiMaterialCll();
62     mat.setId(1);
63     mat.setProperties("Default1222", 29000, 0.3, 450);
64     listMat.Add(mat);
65     solver.setMaterials(listMat);
66
67     // define sections
68     List<cgiSectionCll> listSect = new List<cgiSectionCll>();
69     cgiSectionCll sect1 = new cgiSectionCll();
70     sect1.setId(1);
71     sect1.setProperties("W27X84", 24.8, 12.282, 12.7488, 2850, 106, 2.81);
72     listSect.Add(sect1);
73     cgiSectionCll sect2 = new cgiSectionCll();
74     sect2.setId(2);
75     sect2.setProperties("W27X84", 24.8, 12.282, 12.7488, 2850, 106, 2.81);
76     listSect.Add(sect2);
```

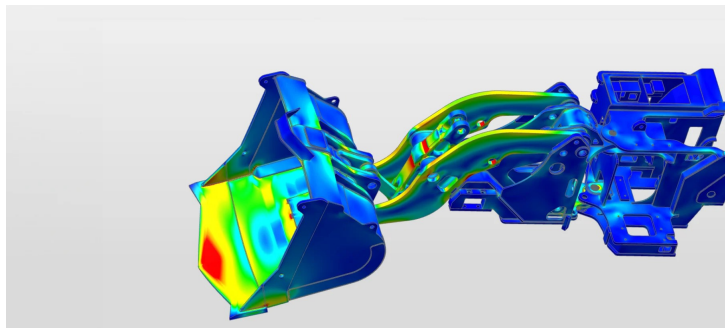
(Source: Cg-inc)



(Source: Padt)



(Source: Ansys)



(Source: Rfamec)

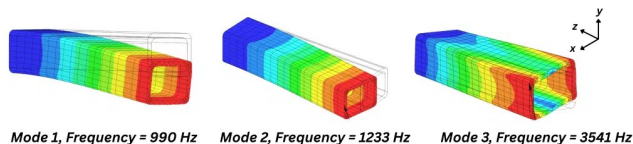
Static analysis determines deformations due to constant loads.

- No matter how you apply the load, the solver assumes it happens infinitely slowly, meaning the loading method has no impact on structural behaviour.
- A constant load doesn't mean a linear response, buckling, yielding, and other nonlinear effects can occur, but the analysis remains static.

The decisive dynamic effect is the presence of inertia forces, forces that arise when the state of motion of a mass changes

Modal analysis studies dynamic properties in the frequency domain.

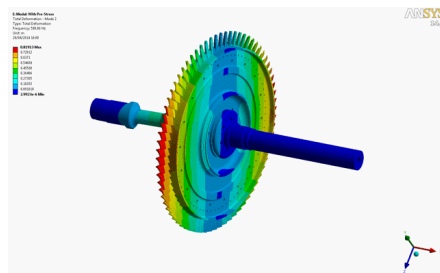
- Natural frequencies are fundamental property in structural design; lower natural frequencies lead to larger displacement and thus greater risk.
- Resonance occurs when external excitation coincides with a natural frequency and can cause catastrophic failure.



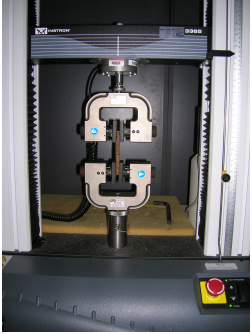
(Source: [CaeFlow](#))

Transient analysis examines time-dependent structural responses to time-dependent excitations.

- Transient analysis helps predict peak stresses, displacements, and vibrations during short-term dynamic events such as impacts, blasts, or sudden load changes.



(Source: [Trivista](#))



(Source: [Wikipedia](#))



(Source: [Wikipedia](#))



(Source: [Wikipedia](#))

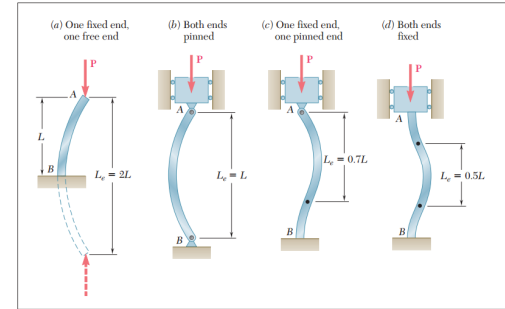


Fig. 10.17 Effective length of column for various end conditions.

(Source: [Engineering Stackexchange](#))



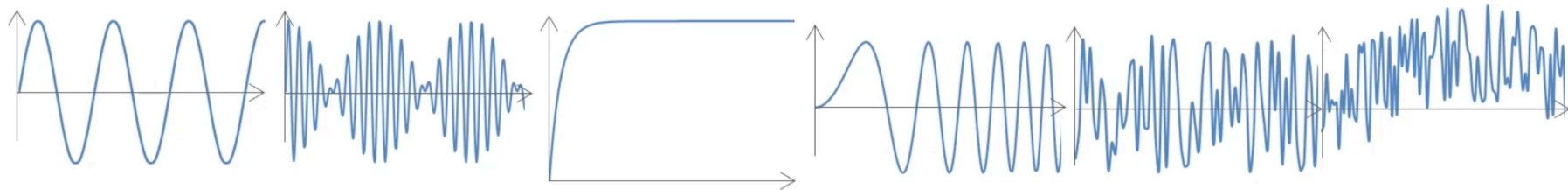
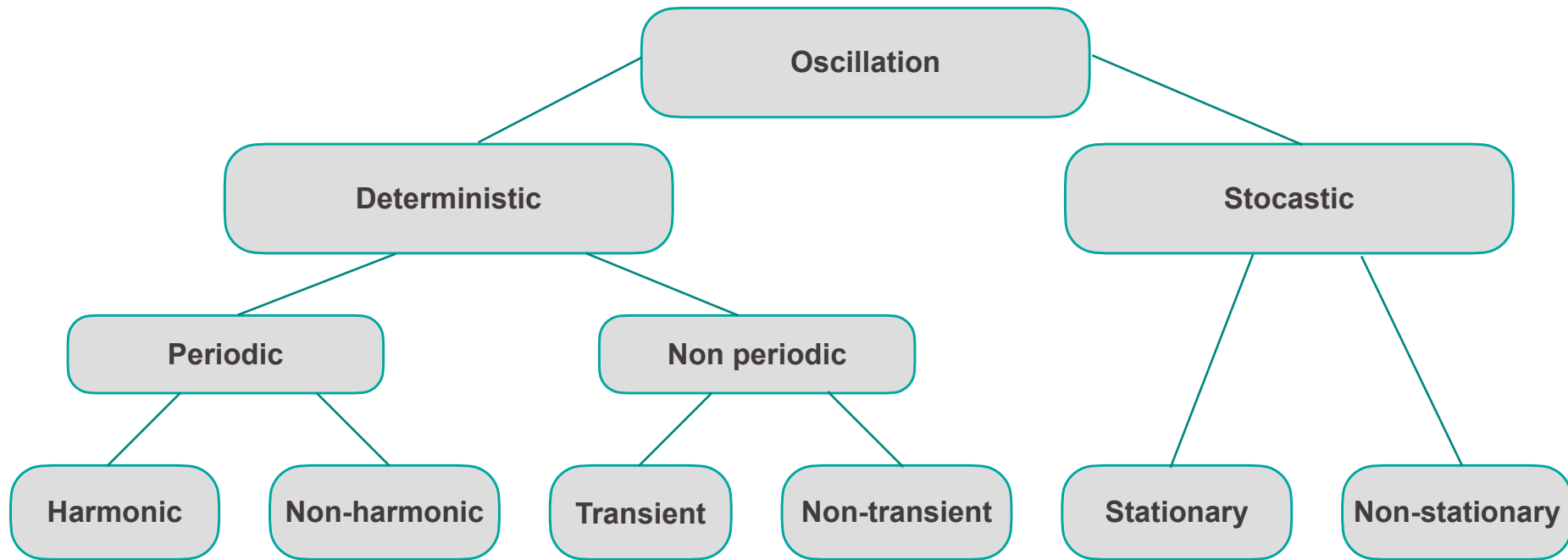
(Source: [Wikipedia](#))



(Source: [Wikipedia](#))



(Source: [Wikipedia](#))

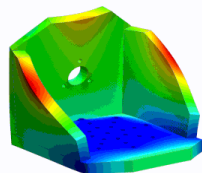


Key distinction: whether the structural stiffness remains constant as the load changes.

Linear analysis

Constant stiffness

$$\mathbf{K} \mathbf{u} = \mathbf{F}$$



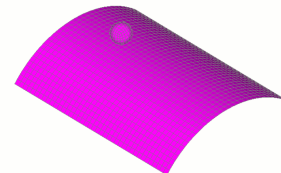
(Source: Sentek)

- Force–displacement relationship is linear.
- Stiffness matrix $\mathbf{K}$ is constant during the analysis.
- Typically solved with one linear system for a load case.
- Suitable when deformations and material response remain linear.

Non linear analysis

Stiffness changes along the load path

$$\mathbf{R}(\mathbf{u}) = \mathbf{F}$$



(Source: Enterfea)

- Force–displacement relationship is not a straight line.
- Tangent stiffness $\mathbf{K}_t$ varies with the current state.
- Requires incremental / iterative solution procedures.
- Common causes: plasticity, large deformation, and changing contact or element status.

